

VEGETATION CHANGE DETECTION IN KATHMANDU VALLEY USING NDVI TIMELAPSE (2000–2026)

1. INTRODUCTION

This project analyzes vegetation change in Kathmandu Valley from 2000 to 2026 using NDVI and satellite imagery. Landsat 5, Landsat 7, Landsat 8, and Sentinel-2 data were used to create yearly NDVI timelapse maps.

2. OBJECTIVE

To detect vegetation expansion and contraction in Kathmandu Valley from 2000 to 2026 using NDVI analysis.

3. NDVI

NDVI (Normalized Difference Vegetation Index) measures vegetation health using red and near-infrared (NIR) light.

$$\text{NDVI} = (\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red})$$

Higher NDVI values indicate healthier vegetation.

4. NDVI CLASSIFICATION

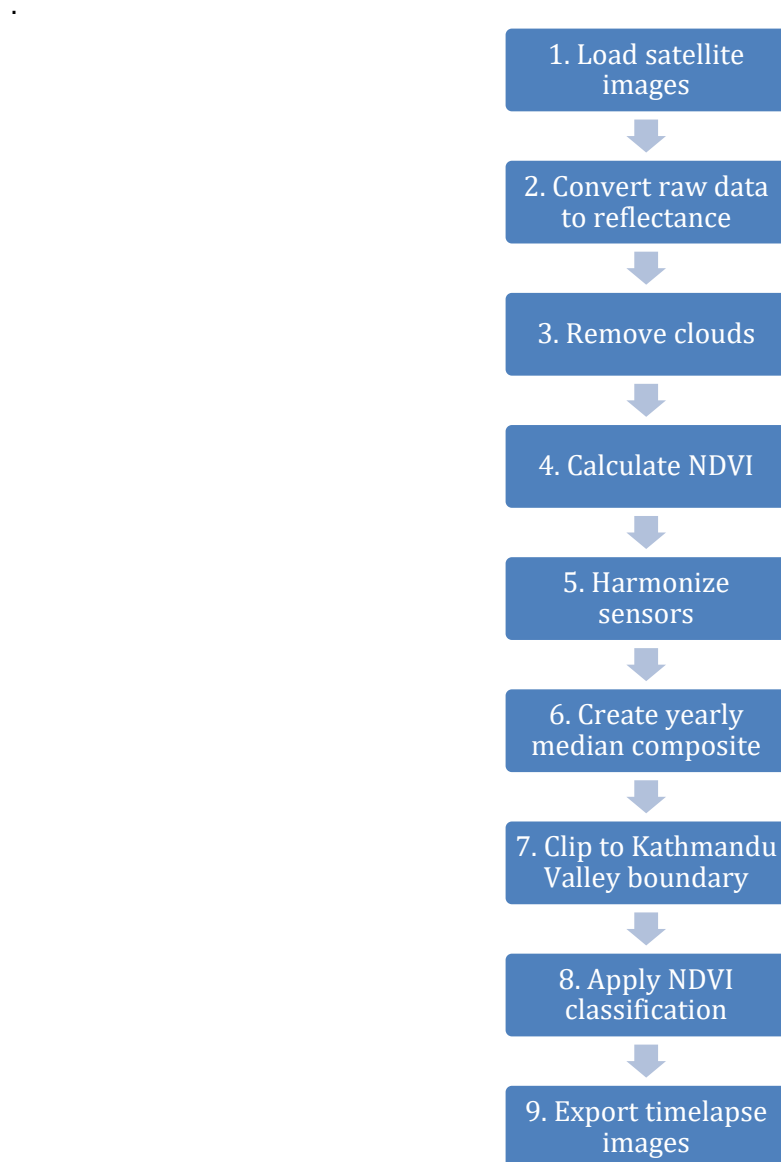
- -0.1 to 0.1 → Built-up area (Red)
- 0.1 to 0.25 → Very thin vegetation (Orange)
- 0.25 to 0.4 → Low vegetation (Yellow)
- 0.4 to 0.6 → Light forest (Light Green)
- 0.6 to 1.0 → Dense forest (Dark Green)

5. DATA SOURCES

- Landsat 5 (2000–2011) – 30 m
- Landsat 7 (2002 only) – 30 m
- Landsat 8 (2013–2015) – 30 m
- Sentinel-2 (2015–2026) – 10 m

6. METHODOLOGY

Workflow:



7. CLOUD MASKING

Clouds were removed using:

- SCL band for Sentinel-2
- QA_PIXEL band for Landsat

8. SENSOR HARMONIZATION

Different sensors produce slightly different NDVI values. Harmonization formulas were applied to make all sensors comparable.

9. MEDIAN COMPOSITE

Median composite was used to remove remaining clouds and generate one clean image for each year.

10. MANUAL VS AUTOMATED WORKFLOW

Manual ArcGIS Pro:

- Processing Time: 6–7 hours
- Advantages: Full control and easy visualization
- Limitations: Time consuming and repetitive

Google Earth Engine:

- Processing Time: 10–15 minutes
- Advantages: Fast, scalable, cloud-based
- Limitation: Requires coding knowledge

11. RESULTS

The timelapse shows:

- Urban expansion inside Kathmandu Valley
- Stable forest cover in hill regions
- Vegetation loss in built-up areas
- Local vegetation recovery in some regions

12. CONCLUSION

This project successfully created a 26-year NDVI timelapse using Landsat and Sentinel satellite imagery. The workflow included cloud masking, NDVI calculation, sensor harmonization, and annual median composite generation. Google Earth Engine significantly reduced processing time compared to manual GIS processing.

13. Limitations

LIMITATIONS

1. Missing Data Years (2002 and 2012)
Landsat 5 failed in early 2002. No usable data for 2002. Landsat 5 decommissioned November 2011. Landsat 8 launched February 2013. Gap of 12-14 months with no data for 2012.
2. Sensor Harmonization Needed
Landsat and Sentinel-2 measure different wavelength ranges. Same forest gives different NDVI values on different sensors. Required mathematical correction. Small systematic differences remain after harmonization.
3. Cloud Cover
Monsoon season clouds remove many pixels. Some areas remain as NoData holes in final composite.
4. Resolution Difference
Landsat is 30 meters. Sentinel-2 is 10 meters. Apparent detail increase after 2015 not actual vegetation change.
5. Seasonal Differences
Images acquired in different months. April shows different greenness than October. Year-to-year fluctuations may be seasonal, not long-term change.
6. Topographic Shadow
Hills create shadows. North-facing slopes show lower NDVI than south-facing slopes with same vegetation.
7. Urban vs Bare Soil Confusion
Freshly plowed fields have NDVI near zero. Sometimes classified as built-up area incorrectly.
8. Atmospheric Correction Not Perfect
Residual haze or dust may remain. Affects NDVI values, especially in lower range.

14. CODES

```
// SENTINEL-2: Reference standard (no harmonization needed)
// Why: Sentinel-2 Level-2A is already atmospherically corrected
// Why: Divide by 10000 converts to reflectance (0-1 range)
// Why: SCL band values 4,5,6,7 = good pixels (vegetation, non-vegetated, water, unclassified)

var boundary = ee.FeatureCollection('projects/ee-abhishekkarki520/assets/Boundary');
Map.centerObject(boundary, 10);

function maskS2(image) {
  var scl = image.select('SCL');
  var good = scl.eq(4).or(scl.eq(5)).or(scl.eq(6)).or(scl.eq(7));
  return image.updateMask(good);
}

var year = 2020; // CHANGE YEAR

var s2 = ee.ImageCollection('COPERNICUS/S2_SR_HARMONIZED')
  .filterBounds(boundary)
  .filterDate(year + '-01-01', year + '-12-31')
  .filter(ee.Filter.lt('CLOUDY_PIXEL_PERCENTAGE', 50))
  .map(maskS2);

var ndvi = s2.map(function(img) {
  var nir = img.select('B8').divide(10000);
  var red = img.select('B4').divide(10000);
  return nir.subtract(red).divide(nir.add(red)).rename('NDVI');
}).median().clip(boundary);

Map.addLayer(ndvi, {min: -0.2, max: 0.8, palette: ['red', 'orange', 'yellow', 'lightgreen', 'darkgreen']}, 'S2 ' + year);

Export.image.toDrive({
  image: ndvi,
  description: 'NDVI_S2_' + year,
  folder: 'NDVI_Timelapse',
  scale: 10,
  region: boundary.geometry(),
  maxPixels: 1e9
});
```

```

// LANDSAT 8: Harmonized to match Sentinel-2
// Why: Scale by 0.0000275 -0.2 converts to reflectance
// Why: Multiply by 0.965 + 0.026 harmonizes to Sentinel-2 (Claverie et al. 2018)
// Why: QA_PIXEL bits 3 and 4 remove clouds

var boundary = ee.FeatureCollection('projects/ee-abhishekkarki520/assets/Boundary');
Map.centerObject(boundary, 10);

function maskL8(image) {
  var qa = image.select('QA_PIXEL');
  var mask = qa.bitwiseAnd((1 << 4)).eq(0).and(qa.bitwiseAnd((1 << 3)).eq(0));
  return image.updateMask(mask);
}

var year = 2014; // CHANGE YEAR

var l8 = ee.ImageCollection('LANDSAT/LC08/C02/T1_L2')
  .filterBounds(boundary)
  .filterDate(year + '-01-01', year + '-12-31')
  .filter(ee.Filter.lt('CLOUD_COVER', 50))
  .map(maskL8);

var ndvi = l8.map(function(img) {
  var nir = img.select('SR_B5').multiply(0.0000275).add(-0.2);
  var red = img.select('SR_B4').multiply(0.0000275).add(-0.2);
  var raw = nir.subtract(red).divide(nir.add(red));
  return raw.multiply(0.965).add(0.026).rename('NDVI');
}).median().clip(boundary);

Map.addLayer(ndvi, {min: -0.2, max: 0.8, palette: ['red', 'orange', 'yellow', 'lightgreen', 'darkgreen']}, 'L8 ' + year);

Export.image.toDrive({
  image: ndvi,
  description: 'NDVI_L8_' + year,
  folder: 'NDVI_Timelapse',
  scale: 30,
  region: boundary.geometry(),
  maxPixels: 1e9
});

```

```

// LANDSAT 5: Harmonized to match Sentinel-2
// Why: Scale by 0.0000275 -0.2 converts to reflectance
// Why: Multiply by 0.987 + 0.022 harmonizes to Sentinel-2 (Vermote et al. 2016)
// Why: SR_B3 = Red, SR_B4 = NIR for Landsat 5

var boundary = ee.FeatureCollection('projects/ee-abhishekkarki520/assets/Boundary');
Map.centerObject(boundary, 10);

function maskL5(image) {
  var qa = image.select('QA_PIXEL');
  var mask = qa.bitwiseAnd((1 << 4)).eq(0).and(qa.bitwiseAnd((1 << 3)).eq(0));
  return image.updateMask(mask);
}

var year = 2000; // CHANGE YEAR

var l5 = ee.ImageCollection('LANDSAT/LT05/C02/T1_L2')
  .filterBounds(boundary)
  .filterDate(year + '-01-01', year + '-12-31')
  .filter(ee.Filter.lt('CLOUD_COVER', 50))
  .map(maskL5);

var ndvi = l5.map(function(img) {
  var nir = img.select('SR_B4').multiply(0.0000275).add(-0.2);
  var red = img.select('SR_B3').multiply(0.0000275).add(-0.2);
  var raw = nir.subtract(red).divide(nir.add(red));
  return raw.multiply(0.987).add(0.022).rename('NDVI');
}).median().clip(boundary);

Map.addLayer(ndvi, {min: -0.2, max: 0.8, palette: ['red', 'orange', 'yellow', 'lightgreen', 'darkgreen']}, 'L5 ' + year);

Export.image.toDrive({
  image: ndvi,
  description: 'NDVI_L5_' + year,
  folder: 'NDVI_Timelapse',
  scale: 30,
  region: boundary.geometry(),
  maxPixels: 1e9
});

```

```

// LANDSAT 7: Only for 2002 (Landsat 5 failed this year)
// Why: Same bands as Landsat 5 (SR_B3=Red, SR_B4=NIR)
// Why: Harmonization: multiply by 0.992 + 0.018 (Teillet et al. 2001)

var boundary = ee.FeatureCollection('projects/ee-abhishekkarki520/assets/Boundary');
Map.centerObject(boundary, 10);

function maskL7(image) {
  var qa = image.select('QA_PIXEL');
  var mask = qa.bitwiseAnd((1 << 4)).eq(0).and(qa.bitwiseAnd((1 << 3)).eq(0));
  return image.updateMask(mask);
}

var l7 = ee.ImageCollection('LANDSAT/LE07/C02/T1_L2')
  .filterBounds(boundary)
  .filterDate('2002-01-01', '2002-12-31')
  .filter(ee.Filter.lt('CLOUD_COVER', 50))
  .map(maskL7);

var ndvi = l7.map(function(img) {
  var nir = img.select('SR_B4').multiply(0.0000275).add(-0.2);
  var red = img.select('SR_B3').multiply(0.0000275).add(-0.2);
  var raw = nir.subtract(red).divide(nir.add(red));
  return raw.multiply(0.992).add(0.018).rename('NDVI');
}).median().clip(boundary);

Map.addLayer(ndvi, {min: -0.2, max: 0.8, palette: ['red', 'orange', 'yellow', 'lightgreen', 'darkgreen']}, 'L7 2002');

Export.image.toDrive({
  image: ndvi,
  description: 'NDVI_L7_2002',
  folder: 'NDVI_Timelapse',
  scale: 30,
  region: boundary.geometry(),
  maxPixels: 1e9
});

```